MIL-A-18065A(SHIPS)
23 June 1966

SUPERSEDING
MIL-A-18065(SHIPS)
22 July 1954
(See 6.5)

MILITARY SPECIFICATION

ADHESIVES, HIGH INITIAL BOND

1. SCOPE

1.1 Scope. - This specification covers high-initial bond adhesive for use in securing elastomeric plastic foam insulation (MIL-P-15280) to metal surfaces in submarines.

1.2 Classification. - Adhesive shall be of the following classes, as specified (see 6.2):

Class 1 - Fast-setting
Class 2 - Slow-setting

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
PPP-C-96 - Cans, Metal, 28 Gage and Lighter.
PPP-P-704 - Pails: Shipping, Steel (1 Through 12 Gallon).

MILITARY
MIL-P-15280 - Plastic Foam, Unicellular, Sheet and Tubular Form, Elastomeric.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or reque for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D92 - Method of Test for Flash and Fire Points by Cleveland Open Cup.
ASTM D217 - Method of Test for Cone Penetration of Lubricating Grease.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

FSC 8040

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OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd St., New York, N.Y. 10016)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Qualification. - Adhesive furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.3).

3.2 Material. - Adhesive, in the condition in which delivered, shall be suitable and effective for the purpose intended without heating or additions of other ingredients, and shall be free from grit, lumps, and skins, and shall not gel, liver, settle hard or otherwise deteriorate when stored for a period of one year in airtight containers. The adhesive shall be free from all ingredients which may effect the serviceability or have a deleterious effect on the metal or insulation to which the adhesive is applied. The adhesive shall be free from chlorinated solvents, benzol, or any other toxic ingredients which would adversely affect the health of workmen using them.

3.3 Protection against corrosion. - Adhesive shall protect bare steel surfaces from corrosion when tested as specified in 4.5.1.

3.4 Flexibility. - Adhesive shall not chip, scale, crack or otherwise leave the surface of the tin strip when tested as specified in 4.5.2.

3.5 Consistency. - The cone penetration of adhesive shall be not less than 25 or more than 35 millimeters (mm.) (see 4.5.3).

3.6 Shear strength. - The shear strength of adhesive shall be sufficient to support a one-pound weight for a period of 72 hours (see 4.5.4).

3.7 Initial bond adhesive strength. - The initial bond adhesive strength of the adhesive shall be not less than 60 pounds per square foot (see 4.5.5).

3.8 Class 1. -

3.8.1 General requirements. - Adhesive shall be a fast-setting, high-initial bond, volatile-solvent-type adhesive suitable for securing elastomeric plastic foam sheet conforming to MIL-P-15280 to bare steel surfaces in easily ventilated compartments (see 6.1).

3.8.2 Ultimate adhesive strength. -

3.8.2.1 The average ultimate adhesive strength shall be not less than the values specified in table I.

Table I - Ultimate adhesive strength.

	Method of test, paragraph			
	4.5.6.3.1	4.5.6.3.2	4.5.6.3.3	4.5.6.3.4
Ultimate adhesive strength, pounds per square foot, minimum	1,500	1,000	1,500	1,000

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3.8.2.2 Adhesive shall be considered as conforming to the strength requirements specified in 3.8.2.1 when failure occurs in the insulation material at a strength lower than specified in table I for the adhesive.

3.9 Class 2. -

3.9.1 General requirements. - Adhesive shall be a nonhardening, high-initial-bond, high-flash point, nontoxic-solvent-type adhesive suitable for securing elastomeric plastic foam conforming to MIL-P-15280 to bare steel surfaces in confined, hard-to-ventilate compartments (see 6.1).

3.9.2 Ultimate adhesive strength. -

3.9.2.1 The average ultimate adhesive strength shall be not less than the values specified in table II.

Table II - Ultimate adhesive strength.

	Method of test, paragraph			
	4.5.6.3.1	4.5.6.3.2	4.5.6.3.3	4.5.6.3.4
Ultimate adhesive strength, pounds per square foot, minimum	1,500	250	750	750

3.9.2.2 Adhesive shall be considered as conforming to the strength requirements specified in 3.9.2.1 when failure occurs in the insulation material at a strength lower than that specified in table II for the adhesive.

3.9.3 Flash point. - The flash point of the adhesive shall be not less than 120° Fahrenheit (F.) (see 4.5.7).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification tests. [1/] - Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. Qualification tests shall consist of the test specified in 4.5.

4.3 Sampling for inspection. -

4.3.1 Lot. - For purposes of inspection, a lot shall consist of all adhesive from one production batch offered for delivery at one time.

4.3.2 Sampling for examination of filled containers. - A random sample of filled containers shall be selected from each lot in accordance with MIL-STD-105 at inspection level S1 and acceptable quality level equal to 2.5 percent defective for the examination specified in 4.4.1.

4.3.3 Sampling for inspection tests. From each lot, two containers shall be selected. From each of the containers, two-quart samples shall be taken for the tests specified in 4.4.2.

[1/] Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3 and 6.4).

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4.4 Inspection. -

4.4.1 Examination of filled containers. - Each sample filled container selected in accordance with 4.3.2 shall be examined to verify compliance with this specification. Any container in the sample having one or more defects or under required fill shall be cause for rejection, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, it shall be cause for rejection of the lot which it represents.

4.4.2 Inspection tests. - The sample specimens selected in accordance with 4.3.3 shall be separately subjected to the tests specified in 4.5. If any one of the samples tested is found to be not in conformance with this specification, it shall be cause for rejection of the lot which it represents.

4.5 Test procedures. -

4.5.1 Protection against corrosion. - Ground steel strips 1/8 by 1 by 6 inches shall be dipped in the adhesive and allowed to dry for 24 hours. At the end of the time the strips shall be subjected to a 4-percent salt solution spray for 72 hours at 100°F. The strips shall be removed from the spray and examined for penetration of the salt spray and corrosion of the metal surfaces.

4.5.2 Flexibility. -

4.5.2.1 Apparatus. - The apparatus shall consist of a 8 by 12 by 0.011 inch panel of bright tinplate, a film applicator with a clearance of 0.02 inch and a gap of sufficient size to make a film 3 to 6 inches wide, and a steel mandrel 1/4 inch in diameter and 6 inches long supported at each end.

4.5.2.2 Procedure. - The panel shall be placed on a flat surface and a film of the adhesive 12 inches long and 3 to 6 inches wide shall be drawn down using the film applicator. The film shall be allowed to dry at room temperature (77° ± 5°F.) for 24 hours with the panel in horizontal position. Four 1-1/2 by 4 inch strips shall be cut from the coated panel. Two of the strips shall be placed in an air-circulating oven at 212° to 220°F. for 75 minutes. The other two strips shall be immersed in fresh water at a temperature of 75° ± 5°F. for 48 hours. The strips shall be removed from the oven and the water, allowed to dry and cool to room temperature, and then bent rapidly through a 180° arc over the surface of the mandrel. Then the strips shall be inspected for chips, scales and cracks at the bend. Slight surface cracking shall be disregarded, but deep cracks which expose bare metal shall be considered as failure to pass the test.

4.5.3 Consistency. -

4.5.3.1 Apparatus. - The apparatus used shall be as specified in ASTM D217. The total weight of the cone and rod shall be 150 grams.

4.5.3.2 Procedure. - A thin film of water shall be floated on top of the test adhesive during the test to prevent nonconformity due to evaporation or skinning. The open cup portion of the grease worker as specified in ASTM D217 shall be completely filled with the well-mixed sample at a temperature of 77° ±1°F. A straight edge shall be drawn across the surface of the sample to remove any excess material and to present a smooth surface. The apparatus shall be leveled and the plunger lowered until the tip of the penetrometer cone just touches the surface of the sample. The scale shall then be adjusted so that the scale actuating device is in contact with the top of the rod holding the penetrometer cone and the scale reading recorded. The plunger shall be released suddenly and kept released for 5 seconds. The scale actuating device shall be moved until it is again in contact with the top of the rod holding the penetrometer cone, and the scale reading recorded. The penetration is the difference between the two readings. Five tests shall be made and the average reported. The sample shall be smoothed over before each test.

4.5.4 Shear strength. - A 6 by 6 by 1-inch section of elastomeric plastic foam insulation conforming to MIL-P-15280 shall be trowel-coated on one face with the adhesive and allowed to set for 5 minutes. The insulation then shall be applied with a rotary sliding motion to the vertical, mill scale surface of a steel plate. A 1-pound weight shall be immediately suspended from the specimen with a light sheet metal yoke framed around the insulation adjacent to the adhesive line. The specimen shall be observed closely for slippage of the material and shear of the cement. If no initial shear occurs, the specimen shall be observed over a 72-hour period for possible failure.

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4.5.5 Initial-bond adhesive strength. -

4.5.5.1 Apparatus. - The apparatus shall consist of two aluminum discs 7-1/2 by 5-7/16 inches in diameter, by 1/4 inch thick, an aluminum ring 5-9/16 diameter by 7-1/2 outer diameter by 1/4 inch thick, and two threaded studs with wingnuts to fit holes drilled in the larger disc and ring to hold them centered, parallel and 1 inch apart.

4.5.5.2 A 6 by 6 by 1-inch specimen of elastomeric plastic foam insulation conforming to MIL-P-15280 shall be trowel-coated on one side with the test adhesive and laid coated side up on the larger disc. The ring then shall be bolted in place on the coated side, and the assembly shall be dried at room temperature (77° ±5°F.) for 5 minutes. Then the smaller disc shall be placed in the hollow of the ring, its center concentric to that of the ring, using a rotary sliding motion to insure full contact between the disc and the adhesive. Hooks shall be secured in centered holes on the outer faces of the two discs. The adhesion of the smaller disc to the adhesive shall be determined immediately in a tensile machine, the lower jaw of which shall move at a uniform rate of 12 ±1/2 inches per minute. Initial bond adhesive strength shall be the average of three determinations.

4.5.6 Ultimate adhesive strength. -

4.5.6.1 Apparatus. - Two steel discs 4 inches in diameter and 1/2 inch thick shall be drilled and tapped in the center to receive a 1/2 inch bolt. The tapped holes shall not be drilled entirely through the discs. Each bolt shall be drilled radially at the outer end to receive a swivel joint steel pin for securing to the testing machine.

4.5.6.2 Preparation. - The adhesive shall be applied with a serrated trowel to the surface of the two steel discs and allowed to set for 5 minutes. The coated steel discs shall then be applied to the two surfaces of a 4-inch elastomeric plastic foam disc, 1 inch thick, in such a manner that the elastomeric plastic foam disc will be sandwiched between the steel discs.

4.5.6.3 Procedure. -

4.5.6.3.1 Three specimens shall be prepared as specified in 4.5.6.2. They shall be dried for 72 hours at room temperature (77° ±5°F.). Then they shall be pulled to rupture in a testing machine, the lower jaw of which moves at a uniform rate of 12 ±1/2-inch per minute.

4.5.6.3.2 Three specimens shall be prepared as specified in 4.5.6.2. They shall be dried for 72 hours at room temperature (77° ±5°F.) and for 2 hours at 158°F. Then they shall be pulled to rupture in a testing machine, the lower jaw of which moves at a uniform rate of 12 ±1/2 inch per minute.

4.5.6.3.3 Three specimens shall be prepared as specified in 4.5.6.2. They shall be dried for 72 hours at room temperature and immersed in tap water for 24 hours. Then they shall be pulled to rupture in a testing machine, the lower jaw of which moves at a uniform rate of 12 ±1/2-inch per minute.

4.5.6.3.4 Three specimens shall be prepared as specified in 4.5.6.2. The adhesive and plastic foam discs shall be cooled to 20°F. for 24 hours prior to preparation of the specimens. The specimens shall be prepared singly and rapidly to avoid condensation of moisture on the steel discs. They shall be dried for 72 hours at 20°F. Then they shall be pulled to rupture in a testing machine, the lower jaw of which moves at a uniform rate of 12 ±1/2 inch per minute.

4.5.7 Flash point. - The flash point shall be determined in accordance with ASTM D92.

4.5.8 Storagability. - The adhesive shall be stored for a period of one year in airtight containers. At the end of this period, the adhesive shall be examined to determine conformance with 3.2.

4.6 Preparation for shipment. - Examination for packing and marking for shipment shall be made for conformance with the requirements of Section 5.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. - Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. - The adhesive shall be furnished in 1-gallon multifriction top cans conforming to type V, class 2 of PPP-C-96 or 5-gallon lug cover pails conforming to type II of PPP-P-704.

5.1.2 Level C. - Packaging shall be sufficient to afford adequate protection against physical damage during shipment from the supply source to the using activity and until early use.

5.2 Packing. - Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. -

5.2.1.1 The 1-gallon multifriction top can shall be arranged and packed in accordance with the overseas equipment requirements of the appendix to PPP-C-96.

5.2.1.2 The 5-gallon lug cover pails shall require no packing.

5.2.2 Level B. -

5.2.2.1 The 1-gallon multifriction top cans shall be arranged and packed in accordance with the domestic shipment requirements of the appendix to PPP-C-96.

5.2.2.2 The 5-gallon lug cover pails shall require no packing.

5.2.3 Level C. - Adhesive shall be packed in containers, at the lowest rates, in a manner which will insure acceptance by common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the first receiving activity for immediate use. This level in general shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may be the supplier's commercial practice when such meets the requirements of this level.

5.3 Marking. - In addition to any special marking required, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129 (see 6.2).

6. NOTES

6.1 Class 1 (fast setting) adhesive. - Class 1 (fast-setting) adhesive is intended for use in areas where fire and personnel hazards can be eliminated by ventilation.

6.1.1 Class 2 (slow-setting) adhesive. - Class 2 (slow-setting) adhesive is intended for use in dead air spaces where use of class 1 adhesive would present a hazard.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class required (see 1.2).
(c) Quantity required (see 5.1.1 and 5.1.2).
(d) Selection of applicable level of packaging and level of packing required (see 5.1 and 5.2).
(e) Special markings required (see 5.3).

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in Qualified Products List QPL-18065. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose or offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for qualified products list is the Naval Ship Engineering Center, Department of the Navy, Washington, D. C. 20360, and information pertaining to qualification of products may be obtained

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from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6. 4).

6. 4 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6. 5 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - SH
(Project 8040-N020Sh)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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